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AI Customer Support Reply Evaluator

Final Project Report

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Natural Language Processing

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1. Abstract

An automatic evaluation system was implemented for customer support replies using an instruction-tuned large language model. The system evaluates customer support replies based on a predefined rubric and produces both a numerical score and a rationale explaining the decision.

The system uses a dataset with various customer support scenarios along with the appropriate reference, candidate replies, rubrics for evaluating replies, scores, and rationales assigned by humans. A baseline instruction-tuned model was first tested on the dataset. Next, the model was fine-tuned using LoRA (Low-Rank Adaptation).

Evaluation was carried out using various metrics such as Accuracy, Mean Absolute Error (MAE), and Quadratic Weighted Kappa (QWK). Improvement was observed across all evaluation metrics after fine-tuning. Moreover, a Gradio based UI was created to enable interactive evaluation of customer support replies.

The project showcases how instruction-tuned transformer models can be fine-tuned and used for scoring tasks using rubrics in customer support settings.

2. Introduction

Motivation

Customer support quality has a major impact on customer satisfaction and company reputation. Modern businesses receive a large number of customer complaints and inquiries daily through online platforms, websites, and applications. Evaluating the quality of customer support replies manually can be difficult, time-consuming, and inconsistent.

Human evaluators may assign different scores to the same response depending on personal judgment and interpretation. In addition, manual evaluation becomes impractical when dealing with large-scale customer support systems where thousands of interactions occur every day.

With the recent development of Large Language Models (LLMs), it has become possible to automate many language-related tasks with high accuracy and flexibility. Instruction-tuned transformer models are capable of understanding context, comparing responses against references, and generating meaningful explanations.

This project was motivated by the need for a consistent and explainable automatic evaluation system for customer support replies. Instead of relying only on numerical scores, the system also generates rationales explaining why a particular response received a specific score. This makes the evaluation process more transparent and interpretable.

Background

Large Language Models (LLMs) have recently achieved significant success in various Natural Language Processing (NLP) tasks such as text generation, summarization, translation, sentiment analysis, and question answering. Models such as Mistral, LLaMA, and GPT-based systems have shown strong language understanding and reasoning abilities.

Instruction-tuned models are a specialized type of language model trained to follow structured instructions provided in prompts. These models can perform complex tasks when given carefully designed input instructions.

In this project, the evaluation task follows a rubric-based scoring methodology. Rubric-based evaluation is commonly used in educational grading systems, essay assessment, and customer service quality assurance. The method evaluates responses according to predefined criteria instead of relying on subjective overall impressions.

The project uses the Mistral-7B-Instruct model as the base model. This model was selected because it provides strong instruction-following capabilities while remaining efficient enough to fine-tune using LoRA and 4-bit quantization techniques.

LoRA (Low-Rank Adaptation) is a parameter-efficient fine-tuning method that updates only small adapter layers instead of retraining the entire model. This significantly reduces GPU memory usage and training time while maintaining strong performance improvements.

The final system combines:

- Structured prompt engineering
- Rubric-based evaluation
- Instruction tuning
- LoRA fine-tuning
- Automated rationale generation

to create a complete AI-based customer support evaluation pipeline.

3. Dataset

Source

The dataset used in this project was manually created and curated by the project team. The final dataset contains 200 customer support evaluation samples, where each project member created 100 samples covering a variety of customer support scenarios.

Since no publicly available dataset perfectly matched the required project structure, AI-assisted generation tools were used to help create realistic customer support conversations and responses. However, all generated samples were carefully reviewed, corrected, and refined manually by the project team to ensure quality, consistency, realism, and alignment with the evaluation rubric.

Public customer support examples and online complaint scenarios were used only as inspiration for designing realistic situations. All final dataset entries, including the tasks, reference answers, submissions, rubrics, scores, and rationales, were manually verified by the project team.

The dataset was intentionally designed to simulate real-world customer support interactions across multiple categories of customer issues. These scenarios include order delays, damaged products, refund requests, wrong item deliveries, payment issues, subscription cancellations, website or application problems, account login issues, customer dissatisfaction, and missing packages.

This diversity of scenarios helped improve the model's ability to generalize across different customer support situations and evaluation cases.

Description

The dataset was designed as a structured rubric-based evaluation dataset for customer support replies. Each dataset entry represents a complete customer support evaluation example containing six main components: task, reference, submission, rubric, score, and rationale.

The **task** field describes the customer's issue or complaint, such as delayed orders, damaged products, refund requests, login problems, or billing issues. The **reference** field contains an ideal customer support response written in a professional, empathetic, and solution-oriented manner. The **submission** field contains the candidate reply that should be evaluated by the model.

The **rubric** field includes a fixed set of evaluation criteria used to assess the quality of the customer support reply. The rubric focuses on important aspects of customer support communication, including:

- understanding the customer issue,
- showing empathy,
- providing a correct and helpful solution,
- and maintaining a polite and professional tone.

Each sample also includes a **score** ranging from 0 to 4, representing the overall quality of the submission. Lower scores represent weak or unacceptable responses, while higher scores represent professional and highly effective customer support replies.

In addition, every sample contains a **rationale** field, which provides a short human-written explanation justifying the assigned score according to the rubric criteria.

The dataset was intentionally designed to include examples across all score levels in order to help the model learn the differences between poor, average, and high-quality customer support replies. The responses also vary in tone, detail level, empathy, and solution quality to improve the robustness of the evaluation system.

Preprocessing

Several preprocessing steps were applied before training and evaluation to ensure the dataset was clean, structured, and suitable for instruction tuning.

First, all dataset entries were organized in JSONL format, where each line represents a single evaluation sample. This format simplified loading, processing, and training operations.

Next, the dataset was divided into three subsets: training set, validation set, and test set.

The final split contained 160 training samples, 20 validation samples, and 20 test samples. The test set was kept unseen during training and was used only for final evaluation.

The training set was used for fine-tuning the model, the validation set was used for monitoring performance and selecting the best model configuration, while the test set was reserved only for final evaluation to ensure unbiased results.

Each dataset entry was then converted into an instruction-style prompt. The prompt included the customer support task, the reference response, the candidate submission, and the evaluation rubric. The model was instructed to return its output in a structured JSON format containing a score and a rationale.

Additional preprocessing steps included:

- removing unnecessary spaces and formatting inconsistencies,
- validating score labels,
- checking rubric consistency across samples,
- and reviewing rationales to ensure alignment with the assigned scores.

Since large language models may sometimes generate outputs that are not perfectly structured, regular expression-based parsing methods were implemented to reliably extract valid JSON predictions from the generated responses.

These preprocessing steps helped improve training stability, output consistency, and overall evaluation reliability.

4. Methodology

Approach

This project follows a rubric-based automatic evaluation approach using instruction-tuned large language models. The main idea is to simulate a human evaluator that assesses the quality of customer support replies based on predefined criteria.

The system receives a structured input consisting of four main components: the customer task, a reference answer, the candidate submission, and a rubric defining evaluation criteria. Based on these inputs, the model is required to generate two outputs: a numerical score and a natural language rationale explaining the decision.

The overall workflow of the system can be summarized as follows: first, the input data is formatted into an instruction-style prompt. This prompt clearly defines the evaluation task and instructs the model to respond in a structured JSON format. The model then processes the prompt and generates a prediction that includes both the score and justification.

This approach is designed to ensure that the model does not only assign a score but also provides an explanation aligned with the rubric, making the evaluation process more interpretable and transparent.

The methodology is based on the idea that large language models can learn evaluation behavior when trained on high-quality labeled examples that include both scores and rationales. By exposing the model to many such examples during fine-tuning, it learns to associate specific response qualities (such as empathy, clarity, and correctness) with corresponding score levels.

Overall, the approach transforms the problem from simple text generation into a structured reasoning task, where the model acts as an automated evaluator guided by explicit rubric-based instructions.

Models Used

This project is based on instruction-tuned large language models capable of understanding structured prompts and generating both scores and explanations.

The main base model used in this system is **Mistral-7B-Instruct-v0.2**, which is a transformer-based model specifically optimized for following natural language instructions. This model was selected due to its strong performance in reasoning tasks, good instruction-following ability, and efficiency compared to larger models.

To make training feasible on limited computational resources, a **4-bit quantized version** of the model was used. This significantly reduced GPU memory requirements while maintaining strong performance.

In addition to the base model, **LoRA (Low-Rank Adaptation)** was applied for fine-tuning. Instead of updating all parameters of the model, LoRA introduces small trainable adapter layers that allow efficient adaptation to the customer support evaluation task.

The system includes two main model stages:

- **Baseline Model:** The original instruction-tuned Mistral model evaluated directly on the dataset without fine-tuning.
- **Fine-Tuned Model:** The same base model after being fine-tuned on the custom dataset using LoRA.

Both models were used for comparison in order to measure the impact of fine-tuning on evaluation performance and consistency.

Overall, the combination of Mistral-7B-Instruct and LoRA provides a powerful yet efficient setup for building a rubric-based evaluation system with limited hardware resources.

Training Strategy

The training process in this project was designed to efficiently adapt a large instruction-tuned language model to a rubric-based evaluation task while minimizing computational cost.

The system uses LoRA (Low-Rank Adaptation) as the main fine-tuning strategy. Instead of updating all parameters of the base model, LoRA introduces lightweight trainable adapter layers into selected transformer modules. This allows the model to learn task-specific patterns while keeping the majority of the original model weights frozen.

The training process was conducted using the following strategy:

First, the dataset was split into training, validation, and test sets. The training set was used to update the model weights through LoRA fine-tuning. The validation set was used during training to monitor performance and compare different experiment configurations. The test set was kept completely unseen and was only used for final evaluation after training was completed.

Several experiments were conducted by modifying hyperparameters such as learning rate, LoRA rank, number of epochs, and target modules. Each experiment produced a slightly different model version, and their performance was compared using validation metrics.

The best-performing model was selected based on the experiment comparison. Experiment 5 achieved the strongest evaluation results among the tested LoRA configurations, especially in terms of accuracy, MAE, and QWK.

Experiment 5 applied LoRA to more target modules, including q_proj, v_proj, k_proj, and o_proj. This gave the model more flexibility to adapt to the rubric-based evaluation task.

Finally, the best model was saved and later used for inference in the evaluation system and the Gradio user interface.

Evaluation Criteria

The evaluation system in this project is designed to assess how well the model predicts both the quality score and the corresponding rationale for customer support replies.

The model outputs are evaluated using a combination of quantitative and qualitative methods to ensure a comprehensive analysis of performance.

1. Quantitative Evaluation

The main quantitative metrics used in this project are:

- **Accuracy:** Measures the percentage of cases where the predicted score exactly matches the human-annotated score. This provides a simple measure of correctness.
- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and true scores. Since the scoring scale is ordinal (0–4), MAE is important because it captures how far the prediction is from the correct value, not just whether it is correct or incorrect.
- **Quadratic Weighted Kappa (QWK):** Measures the level of agreement between predicted and true scores while taking into account the distance between them. This is particularly important for ordinal classification tasks, where predicting a score of 4 instead of 3 is less severe than predicting 4 instead of 0.

2. Qualitative Evaluation

In addition to numerical metrics, the generated rationales are also evaluated qualitatively. This includes checking:

- whether the explanation is clear and understandable
- whether it correctly reflects the rubric criteria
- whether it aligns with the assigned score
- and whether it is consistent with human judgment.

3. Overall Evaluation Goal

The main goal of the evaluation process is not only to measure prediction accuracy, but also to ensure that the model behaves like a human evaluator by providing consistent, explainable, and rubric-aligned assessments.

This combination of quantitative and qualitative evaluation ensures a more complete understanding of the model's performance in real-world customer support scenarios.

5. Results

Findings

The evaluation results show that the system is capable of learning rubric-based scoring behavior from the training data and generating meaningful rationales for customer support replies.

Before fine-tuning, the baseline instruction-tuned model demonstrated a basic understanding of customer support scenarios. It was able to recognize general issues such as delays, refunds, and product problems. However, its performance was inconsistent in assigning accurate scores according to the rubric. In several cases, the baseline model struggled to clearly distinguish between adjacent score levels, such as 2 and 3 or 3 and 4. Additionally, the generated rationales were sometimes vague and not fully aligned with the rubric criteria.

After applying LoRA fine-tuning on the custom dataset, a noticeable improvement was observed. The fine-tuned model showed better alignment with the rubric definitions and produced more consistent and stable scoring behavior. It also demonstrated improved sensitivity to important evaluation aspects such as empathy, professionalism, and completeness of the response.

Quantitative Improvements

Across the evaluation metrics, improvements were observed after fine-tuning:

- **Accuracy increased**, indicating a higher number of exact score matches with human annotations.
- **Mean Absolute Error (MAE) decreased**, showing that prediction errors became smaller on average.
- **Quadratic Weighted Kappa (QWK) improved**, reflecting stronger agreement between the model and human evaluators, especially for ordinal scoring consistency.

Qualitative Improvements

In addition to numerical improvements, the quality of generated rationales also improved significantly. The fine-tuned model produced explanations that were:

- more aligned with rubric criteria,
- more structured and clear,
- more relevant to the specific customer issue,
- and closer to human-written evaluations.

Model	Accuracy	MAE	QWK
Baseline Model	80.00%	0.20	0.9545
Fine-Tuned LoRA Model (Experiment 5)	90.00%	0.10	0.9744

Detailed Predictions File

A detailed predictions file was also generated for the held-out test set. The file `data/final_predictions.csv` contains one row for each test sample, including the ground-truth score, the baseline model prediction, the fine-tuned model prediction, and the generated rationales. This file was created using `create_final_predictions.py`, which runs inference using the saved fine-tuned LoRA model without retraining.

Discussion

Overall, the results demonstrate that fine-tuning an instruction-tuned language model on a small but well-structured dataset can significantly improve its performance on rubric-based evaluation tasks. The model transitioned from a general-purpose responder to a more specialized evaluator capable of consistent scoring and explanation generation.

However, some limitations still exist. In borderline cases where submissions are very similar in quality, the model may still produce slight inconsistencies in scoring. This is expected due to the subjective nature of customer support quality assessment.

Despite these limitations, the system shows strong potential for automating customer support evaluation tasks in a reliable and explainable manner.

6. Conclusion

This project presented an AI-based system for automatically evaluating customer support replies using instruction-tuned large language models. The system was designed to simulate human evaluation by assigning a score from 0 to 4 based on a predefined rubric, along with a natural language rationale explaining the decision.

A custom dataset of 200 samples was created and carefully curated to represent a wide range of realistic customer support scenarios. The dataset included tasks, reference answers, candidate submissions, rubrics, human-assigned scores, and rationales.

The Mistral-7B-Instruct model was used as the base model and later fine-tuned using LoRA (Low-Rank Adaptation) to adapt it to the specific rubric-based evaluation task. The results showed that fine-tuning significantly improved the model's ability to produce accurate scores and more consistent explanations.

Evaluation using Accuracy, MAE, and QWK demonstrated clear improvements after fine-tuning. In addition, qualitative analysis showed that the model generated more structured and rubric-aligned rationales compared to the baseline model.

Overall, the project successfully demonstrates that large language models can be adapted for structured evaluation tasks in customer support environments, providing both automated scoring and explainable outputs.

Future Work

Although the current system shows strong performance, several improvements can be explored in future work to further enhance its capabilities.

One possible extension is to increase the size and diversity of the dataset. A larger dataset with more varied customer support scenarios would help improve generalization and robustness of the model.

Another improvement is to introduce **multi-lingual support**, especially for Arabic and English customer support interactions, which would make the system more applicable in real-world regional environments.

Future work could also focus on **fine-grained rubric scoring**, where instead of producing a single overall score, the model provides separate scores for each rubric criterion such as empathy, clarity, and correctness. This would increase interpretability and provide deeper feedback.

Additionally, the system could be extended into a **production-level dashboard or API**, allowing integration with real customer support platforms for real-time evaluation of agent responses.

Another interesting direction is incorporating **human feedback loops**, where human evaluators can correct or adjust model predictions, and the system continuously improves over time using reinforcement learning techniques.

Finally, experimenting with different transformer architectures or larger models could further improve evaluation accuracy and reasoning quality.

Overall, these future enhancements would transform the system from a research prototype into a scalable and production-ready evaluation tool.

7. References

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